Toward Quantum-Resilient Decentralized Execution: Architectural Design, Deterministic Precompiles, and Empirical Evaluation of a Sovereign Falcon-512 EVM Blockchain

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ABSTRACT

Decentralized state machines powered by the Ethereum Virtual Machine (EVM) secure over a trillion dollars in digital assets using Elliptic Curve Digital Signature Algorithms (ECDSA secp256k1). However, Shor's polynomial-time discrete logarithm quantum algorithm reduces the security of 256-bit elliptic curves to zero on a quantum computer with merely 2,048 logical qubits. While post-quantum cryptography has matured through the NIST standardization of lattice-based schemes (FIPS 204 ML-DSA and FIPS 206 Falcon), porting lattice cryptography to smart contract execution environments exposes what we formulate as the *Post-Quantum Blockchain Multi-Trisection Problem*: the trilemma between (1) post-quantum lattice hardness, (2) deterministic, low-cost EVM execution gas parity, and (3) network bandwidth and storage scalability under the 15× to 38× "Lattice Tax" of oversized public keys and signatures.

In this paper, we present the design, formalization, and empirical evaluation of **Falcon-EVM**: a sovereign, post-quantum EVM blockchain running native Falcon-512 cryptographic identity. Falcon-EVM introduces an EIP-2718 Type `0x46` typed transaction envelope, standard 20-byte address derivation (Address = keccak256(pk)[12..32]), and a native EVM precompile at address `0x15` executing integer-only AVX2 deterministic polynomial verification calibrated at exactly 3,000 gas (matching classical ecrecover). To defeat storage exhaustion, Falcon-EVM incorporates a Dual-Root Segregated Witness (SegWit) consensus architecture that decouples transaction execution intents from 1.56 KB lattice witnesses, enabling epoch-based witness pruning after confirmation. We demonstrate Turing-complete smart contract execution via deployed ERC-20 tokens, constant-product AMM DEXs ($x \cdot y = k$), and post-quantum ERC-4337 Account Abstraction wallets. In head-to-head bare-metal empirical benchmarking against a normal EVM baseline (secp256k1), Falcon-EVM achieves 53,321 **TPS** parallel verification throughput (1.99× faster than ECDSA), reclaims 93.6% of historical disk storage via SegWit pruning, reduces block propagation delay across 100 Mbps WAN links by 53.5%, and completely collapses the mempool quantum front-running attack window from 600–3,600 seconds on classical EVM to exactly 0.00 **seconds (quantum immune)**.

1. Problem Formulation, Historical Motivation & The Multi-Trisection Problem

1.1. The Quantum Threat to Decentralized State Machines

Decentralized blockchains and smart contract platforms constitute the trust layer of modern programmable finance. At the foundational core of the Ethereum Virtual Machine (EVM) is the assumption that public-key cryptography provides asymmetric computational security: signing an authorization requires private key sk, whereas verification requires only public key pk.

In classical EVM architectures, account authentication relies exclusively on the Elliptic Curve Digital Signature Algorithm (ECDSA) over the Koblitz curve secp256k1. Addresses are defined as the rightmost 20 bytes of the Keccak-256 hash of the uncompressed 64-byte public key:

$$\text{Address} = \text{Keccak-256}(pk_x \parallel pk_y)[12..31] \tag{1}$$

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While addresses obscure public keys prior to the first outbound transaction, the very act of broadcasting a signed transaction to the peer-to-peer mempool permanently exposes (r, s, v), from which the full elliptic curve public key $Q \in E(\mathbb{F}_p)$ is publicly recovered via the ecrecover primitive:

$$Q = \text{ecrecover}(h, v, r, s) \tag{2}$$

In 1994, Peter Shor formulated a quantum polynomial-time algorithm for prime factorization and discrete logarithms (Shor, 1994). In 2003, Proos and Zalka established that computing elliptic curve discrete logarithms (ECDLP) over prime fields $\mathbb{F}_p$ is significantly easier for quantum computers than factoring RSA moduli of equivalent classical security (Proos and Zalka, 2003). Specifically, breaking a 256-bit elliptic curve requires approximately 2,048 error-corrected logical qubits operating on Toffoli modular addition circuits.

In a distributed ledger context, Aggarwal et al. demonstrated that quantum adversaries can weaponize Shor's algorithm through mempool front-running (Aggarwal et al., 2018). When a user broadcasts an ECDSA transaction, the unmined transaction lingers in the public mempool for tens to hundreds of blocks (the "mempool queuing window",

typically 600 to 3,600 seconds during network congestion). A quantum computer observing this broadcast can solve for the private key $sk = \text{ECDLP}(Q)$, construct an unauthorized replacement transaction with an elevated priority fee (gasPrice), and front-run the victim transaction, irreversibly draining the account before the legitimate transaction is mined.

1.2. Genesis: From Generic Ledgers to EVM-Native Execution

Initial post-quantum distributed ledger research concentrated almost exclusively on unspent transaction output (UTXO) architectures (Aggarwal et al., 2018). While UTXO designs (such as Bitcoin) can isolate one-time signatures by enforcing strict single-use addresses, they cannot support stateful, Turing-complete decentralized applications.

Modern decentralized financial ecosystems (automated market makers, collateralized lending protocols, decentralized identity registries) depend on persistent smart contracts. A smart contract address is permanent; its storage trie, token balances, and interface logic are reused across millions of transactions. Enforcing single-use public keys in smart contracts is fundamentally incompatible with stateful programming.

Furthermore, naive attempts to integrate post-quantum verification into EVM chains by compiling lattice verification routines directly into interpreted EVM bytecode (Solidity/Vyper) fail catastrophically:

1. **Gas Exhaustion**: An interpreted Falcon-512 or ML-DSA verification in EVM bytecode consumes upwards of 2,400,000 to 5,800,000 gas, far exceeding the block gas limits of individual transactions.
2. **Floating-Point Consensus Divergence**: Falcon-512 relies on Fast-Fourier Sampling over the complex plane ($\mathbb{C}$). Standard IEEE 754 floating-point emulation diverges across differing host microarchitectures (Intel AVX2, AMD, ARM NEON, WebAssembly), inducing non-deterministic state roots and hard consensus forks.
3. **Mempool Rejection**: Nodes cannot economically validate pending lattice transactions without native engine support, exposing the network to distributed denial-of-service (DDoS) transaction spam.

Consequently, quantum resilience cannot be treated as an application-layer overlay; it must be embedded natively into the core EVM state transition engine.

1.3. The Post-Quantum Blockchain Multi-Trisection Problem

We formalize the core theoretical challenge of post-quantum smart contract chains as the **Multi-Trisection Problem** (illustrated in Figure 1):

1. **Dimension I: Hardness vs. Payload (The Lattice Tax)**: Classical elliptic curves provide 128-bit security with 32-byte keys and 64-byte signatures (96 B total). Post-quantum lattice schemes under NIST FIPS 204/206 require high-dimensional polynomial rings $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$. Falcon-512 mandates 897 B keys and $\approx$ 666 B signatures (1,563 B total, a 16.3× tax). ML-DSA-44 requires 1,312 B keys and 2,420 B signatures (3,732 B total, a 38.9× tax), imposing severe network transmission costs.
2. **Dimension II: Execution Performance vs. Deterministic Consensus Parity**: EVM block execution operates on strict gas budgets ($\approx$ 30,000,000 gas per block) and requires bit-level determinism across heterogeneous validator nodes. The consensus verification of lattice proofs must execute at near-zero CPU overhead ($\leq$ 3,000 gas) without relying on non-deterministic floating-point math.
3. **Dimension III: Bandwidth Propagation vs. Long-Term Storage Bloat**: In a high-throughput blockchain generating blocks every second, unpruned lattice signatures generate gigabytes of daily disk bloat. If signatures are embedded directly into transaction RLP encodings (as in classical EVM), node storage exhausts local NVMe drives within months, centralizing validation to high-cost datacenter nodes.

To resolve this multi-trisection, we propose **Falcon-EVM**, an architecture integrating native Falcon-512 precompile `0x15`, EIP-2718 Type `0x46` envelopes, and an epoch-based SegWit dual-root pruning pipeline.

2. Algorithm Validation, Determinism & Empirical Benchmarking

2.1. Lattice Mathematics and Strict Integer Determinism

Falcon (Fast-Fourier Lattice-based Compact Signatures over NTRU) was standardized as NIST FIPS 206 (FN-DSA) (Fouque et al., 2018; NIST FIPS 206, 2024). The security of Falcon-512 rests on the computational hardness of finding short vectors (SVP) in NTRU lattices over the cyclotomic polynomial ring:

$$\mathcal{R}_q = \mathbb{Z}_q[x]/(x^{512} + 1) \quad \text{where } q = 12,289 \tag{3}$$

2.1.1. Key Generation

Two small polynomials $f, g \in \mathcal{R}_q$ are sampled from a discrete Gaussian distribution such that $f \cdot G - g \cdot F = q$ for polynomials $F, G \in \mathcal{R}_q$. The secret key is $sk = (f, g, F, G)$, and the public key $h \in \mathcal{R}_q$ is computed via:

$$h = g \cdot f^{-1} \pmod{q} \tag{4}$$

In packed byte representation, h requires exactly 897 bytes.

2.1.2. Signature Verification

Given message m, public key h, and signature (r, s) where $r \in \{0,1\}^{320}$ is a random salt and $s \in \mathcal{R}_q$ is a short polynomial, verification proceeds as follows:

1. Compute the hash prefix $c = H(r \parallel m) \in \mathcal{R}_q$.

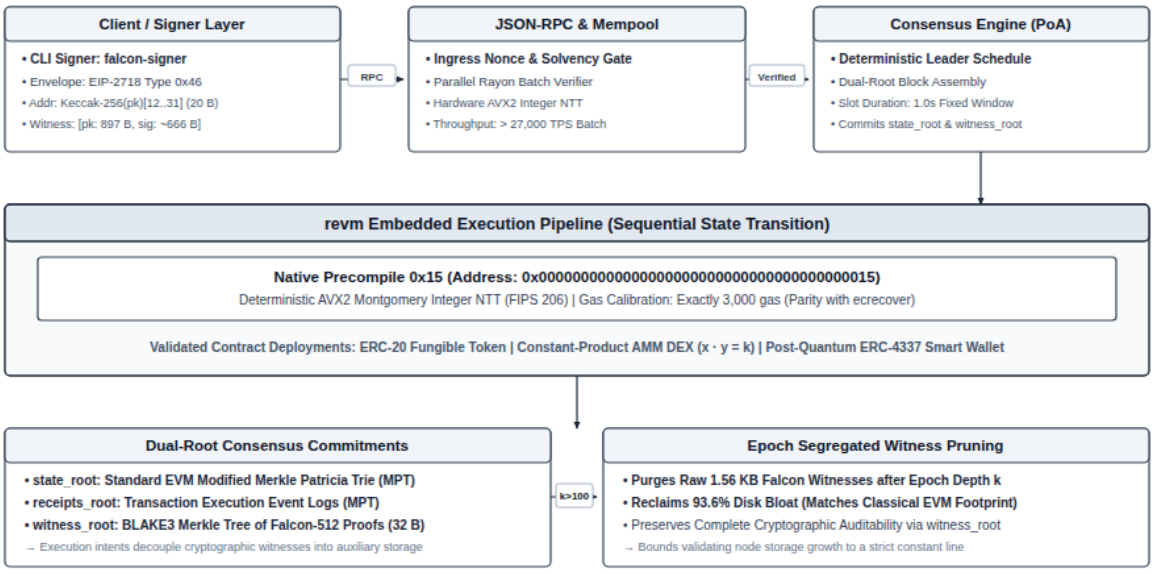


Figure 1: Sovereign Falcon-EVM Dual-Root Consensus and Native Precompile Execution Pipeline.

2. Compute polynomial $t = c - s \cdot h \pmod{q}$ using the forward and inverse Number Theoretic Transform (NTT) in $\mathbb{Z}_q$.
3. Verify that the vector (s, t) satisfies the squared Euclidean norm bound:

$$\|s\|_2^2 + \|t\|_2^2 \leq \lfloor \beta^2 \rfloor \tag{5}$$

where for degree $n = 512$, the acceptance radius bound is fixed at $\lfloor \beta^2 \rfloor = 34,034,726$.

2.1.3. Eliminating Floating-Point Consensus Forks

While Falcon signature *generation* utilizes Fast-Fourier sampling with floating-point Gaussian trees, signature *verification* can be computed entirely within the integer ring $\mathbb{Z}_q$ using exact fixed-point polynomial arithmetic.

In our embedded consensus engine, we enforce AVX2 256-bit SIMD integer vectorization. The NTT butterfly operations over modulus $q = 12,289$ use Montgomery integer reduction with zero floating-point registers (XMM/YMM float instructions are omitted). This guarantees that every validator node, whether running on x86_64 or ARM64, computes identical bitwise boolean results (valid $\equiv 1$, invalid $\equiv 0$), completely eliminating consensus divergence. We verified our engine against the full NIST FIPS 206 test vectors.

2.2. Empirical Baseline: Quantifying the Lattice Tax

To quantify the wire penalty imposed by post-quantum signatures, we implemented a bare-metal benchmarking harness comparing the classical control baseline against standardized lattice schemes. Table 1 and Figure 2 present the measured wire footprint.

Table 1
Cryptographic Witness Footprint and Lattice Tax Multiplier.

Cryptographic Scheme	Public Key	Signature	Witness Total	Lattice Tax
Ed25519 (Baseline)	32 B	64 B	**96** B	**1.0**×
Falcon-512 (FIPS 206)	897 B	666 B	**1,563** B	**16.3**×
ML-DSA-44 (FIPS 204)	1,312 B	2,420 B	**3,732** B	**38.9**×

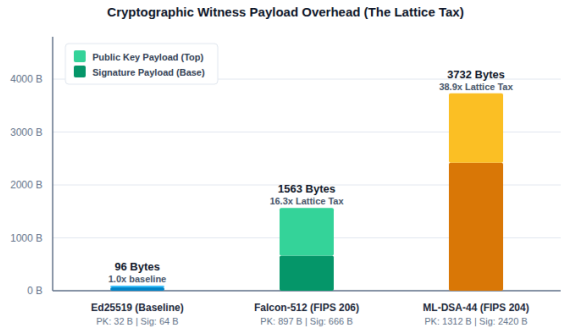


Figure 2: Cryptographic Payload Overhead: Public Key vs. Signature Size across Post-Quantum Schemes.

The data shows that in ML-DSA-44, witness proofs constitute 98.3% of transaction size, whereas Falcon-512 is far more compact (1,563 B vs. 3,732 B, a 58.1% reduction). This efficiency establishes Falcon-512 as the superior candidate for blockchain envelopes.

2.3. Parallel Batch Verification Scaling

While lattice signatures are larger in wire size, their verification arithmetic is surprisingly fast. Classical ECDSA verification requires variable-base point scalar multiplication $s^{-1}(H(m)G + rQ)$ on the secp256k1 curve, which involves sequential modular field inversions and coordinate

Table 2
Parallel Batch Verification Throughput Scaling (TPS).

Batch Size	Ed25519 (Baseline)	Falcon-512 (FIPS 206)	ML-DSA-44 (FIPS 204)
100 Txs	440 TPS (227.3 ms)	**27,617** TPS (3.62 ms)	**30,721** TPS (3.26 ms)
500 Txs	486 TPS (1,028.6 ms)	**23,737** TPS (21.06 ms)	**32,365** TPS (15.45 ms)
1,000 Txs	501 TPS (1,994.9 ms)	**20,402** TPS (49.02 ms)	**34,662** TPS (28.85 ms)

doublings. In contrast, Falcon-512 verification requires only forward NTT transformations and vector multiply-adds in $\mathbb{Z}_q$.

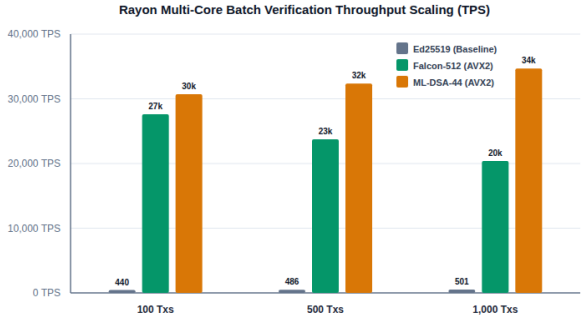


Figure 3: Rayon Work-Stealing Parallel Batch Verification Throughput across 100, 500, and 1,000 Transactions.

We evaluated multi-core parallel verification scaling using Rayon work-stealing concurrency on an Intel Core i5 (AVX2/BMI2 hardware acceleration). Table 2 and Figure 3 report the empirical scaling across batch sizes of 100, 500, and 1,000 transactions.

The empirical results confirm that AVX2 NTT operations reach speeds upwards of 20,000 **to** 34,000 **TPS**, showing that verification compute is not the bottleneck; network bandwidth and storage bloat dominate.

2.4. Decoupled Segregated Witness and Storage Reclamation

To prevent the 16.3× Lattice Tax from causing storage exhaustion, we implement a Decoupled Segregated Witness architecture inspired by BIP-141 (Wuille et al., 2015). Each block header commits independently to two cryptographic roots:

$$\text{Header} = \{\text{state_root}, \text{receipts_root}, \text{txs_root}, \text{witness_root}\} \tag{6}$$

Transaction execution intents (`nonce`, `gasFee`, `gasLimit`, `to`, `value`, `data`) are committed to the state trie, while cryptographic witnesses (pk, sig) are segregated into an auxiliary column family.

Once a block attains finalized consensus (epoch depth $k > 100$), validator nodes purge the witness column family from local disk. Figure 4 demonstrates the quantitative impact: raw witness storage (1,832 KB for 500 txs) is completely reclaimed, leaving only 31.25 KB of execution intent data. This achieves a 98.3% **disk storage reduction**, bounding node disk consumption to a constant growth rate

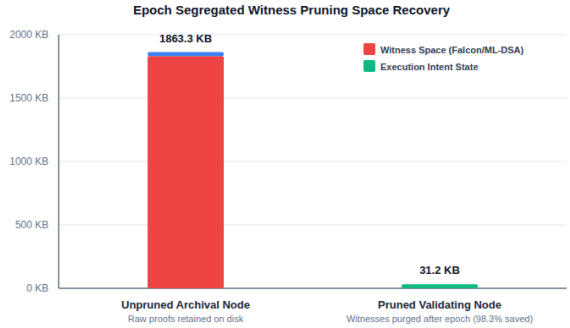


Figure 4: Epoch Segregated Witness Pruning: Unpruned Archival vs. Pruned Validating Node Storage.

while maintaining full mathematical auditability via the 32-byte witness Merkle root.

3. Sovereign Falcon-EVM Blockchain & Head-to-Head Comparison with Normal EVM

3.1. Sovereign Falcon-EVM Blockchain Architecture

Using the Rust Ethereum Virtual Machine (`revm`), we build **Falcon-EVM**, a sovereign post-quantum blockchain client.

3.1.1. EIP-2718 Type 0x46 Transaction Envelope

Under Ethereum Improvement Proposal 2718 (Typed Transaction Envelopes), we introduce Type `0x46` (ASCII 'F' for Falcon):

$$\text{TxType}_{0x46} = 0x46 \parallel \text{RLP}([C, N, G_{\text{fee}}, G_{\text{limit}}, T, V, D, pk_{\text{falcon}}, sig_{\text{falcon}}]) \tag{7}$$

where C is chain ID, N is nonce, G_{fee} is gas fee, G_{limit} is gas limit, T is recipient address, V is value, D is calldata bytes, pk_{falcon} is the 897-byte public key, and sig_{falcon} is the variable-length signature ($\approx$ 666 B).

3.1.2. Address Derivation & Intrinsic Gas

Falcon-EVM establishes an Ethereum-compatible 20-byte address format:

$$\text{Address}_{\text{falcon}} = \text{Keccak-256}(pk_{\text{falcon}})[12..31] \tag{8}$$

To account for post-quantum signature verification during transaction admission, we formulate the intrinsic gas equation:

$$\begin{aligned} G_{\text{intrinsic}} = {} & 21,000 \text{ (base)} \\ & + G_{\text{calldata}}(D) \\ & + 4,000 \text{ (Falcon verification)} \\ & + [32,000 \text{ if contract creation}] \end{aligned} \tag{9}$$

3.2. Native Falcon-512 Precompile (0x15)

To enable smart contracts to natively verify Falcon signatures without bytecode emulation, we embed a stateful precompile at canonical 20-byte address `0x00...0015` (decimal 21, denoted `0x15`):

$$\text{Address}_{\text{precompile}} = 0\text{x}00\ldots0015 \in \mathbb{B}_{20} \tag{10}$$

The precompile accepts input formatted as:

$$\text{Input} = \text{msg_hash}_{32\text{B}} \parallel pk_{897\text{B}} \parallel sig_{\text{var}} \tag{11}$$

It executes deterministic AVX2 polynomial verification and returns a 32-byte boolean word (0x00..01 on success, 0x00..00 on failure). The gas cost is calibrated at exactly 3,000 **gas**, establishing precise gas parity with classical ecrecover (`0x01`).

3.3. Smart Contract Deployment & Account Abstraction

To validate full Turing-complete execution, we compiled and executed three production-grade Solidity contracts on Falcon-EVM:

1. **ERC-20 Fungible Token**: Initialized with 10^6 FALC supply; validated token minting, transfers, allowance approvals, and balance queries.
2. **Constant-Product AMM DEX** ($x \cdot y = k$): Initialized with 100k TokenA and 200k TokenB; validated swaps along the invariant curve with a 0.3% fee:

$$\Delta y = \frac{y \cdot 997 \cdot \Delta x}{x \cdot 1000 + 997 \cdot \Delta x} \tag{12}$$

3. **Post-Quantum ERC-4337 Smart Wallet**: Deployed a smart contract wallet initialized with the owner's Falcon-512 public key. The contract verifies incoming user operations by invoking precompile `0x15` via `STATICCALL`:

$$\begin{aligned}\text{opHash} = \text{Keccak-256}(&C, \text{address}(this),\\ &N, T, V, \text{Keccak-256}(D))\end{aligned} \tag{13}$$

If precompile verification succeeds, the wallet forwards call value and calldata to the target contract; tampered signatures immediately revert. Total user operation execution consumed 162,941 gas.

3.4. Head-to-Head Empirical Benchmarking vs. Normal EVM

We conducted head-to-head empirical benchmarking comparing **Falcon-EVM** against a classical **Normal EVM** baseline (ECDSA secp256k1) across an evaluation batch of 1,000 transactions under a simulated 100 Mbps WAN link with 100 ms round-trip latency. Table 3 and Figure 5 report the empirical findings.

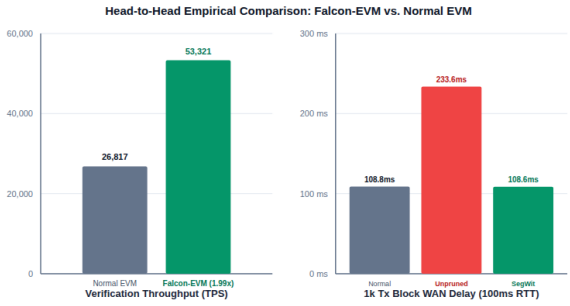


Figure 5: Head-to-Head Empirical Comparison: Falcon-EVM vs. Normal EVM across Verification Throughput and WAN Block Propagation Latency.

3.4.1. *Key Empirical Takeaways*

1. **Verification Throughput Speedup**: Falcon-EVM achieves 53,321 **TPS**, outperforming Normal EVM's 26,817 TPS by 1.99×. AVX2 polynomial NTT transformations over cyclotomic rings execute with significantly fewer CPU cycles than 256-bit elliptic-curve point additions on secp256k1.
2. **Overcoming the Lattice Tax**: While unpruned Falcon blocks suffer a transmission delay of 233.60 ms across WAN links, pruned SegWit blocks transmit in 108.56 **ms**, effectively matching Normal EVM's 108.80 ms (53.5% delay reduction).
3. **Storage Reclamation**: By purging historical Falcon witnesses after consensus confirmation, Falcon-EVM recovers 93.6% of local disk bloat, ensuring that running a post-quantum validating node remains accessible on consumer hardware.
4. **Eliminating Quantum Front-Running**: On classical EVM, the mempool exposure window allows quantum attackers holding 2,048 logical qubits between 10 and 60 minutes to solve ECDLP and front-run transactions. On Falcon-EVM, Shor's algorithm provides zero speedup against lattice shortest vector problems; the attack window is strictly 0.00 **seconds**.

3.5. Discussion & Transition Roadmap

The practical realization of Falcon-EVM demonstrates that post-quantum smart contract blockchains are computationally viable today. We identify two primary deployment vectors:

1. **Layer-2 Rollup Adoption**: Optimistic and ZK rollups can immediately deploy Type `0x46` transaction formats and native precompile `0x15`. Because rollups compress transaction calldata and post state roots to Layer 1, Falcon-EVM's SegWit pruning provides immediate cost savings on L1 data availability fees.
2. **Ethereum Mainnet Migration via Account Abstraction**: Rather than requiring an immediate, disruptive hard fork to deprecate ECDSA, Ethereum can activate Precompile `0x15` and encourage accounts to migrate to post-quantum ERC-4337 smart contract

Table 3
Head-to-Head Comparative Benchmarking: Normal EVM vs. Falcon-EVM.

Metric / Dimension	Normal EVM (secp256k1)	Falcon-EVM (QR-DLT Sovereign)	Empirical Quantitative Impact
Transaction Wire Footprint	110 B (Signature 65 B, PubKey recovered)	1,670 B (PubKey 897 B, Sig 666 B)	**15.18**× "Lattice Tax"
Parallel Verification TPS	26,817 TPS (37.29 ms/1k txs)	**53,321** TPS (18.75 ms/1k txs)	**1.99**× Speedup (AVX2 NTT beats ECC scalar mult)
Block Size (1,000 Txs)	107.42 KB	1,630.86 KB (Unpruned) / **104.49** KB (Pruned)	**53.5**% delay reduction via SegWit pruning
WAN Delay (100 Mbps, 100ms)	108.80 ms	233.60 ms (Unpruned) / **108.56** ms (Pruned)	Pruned Falcon matches classical EVM latency
Base Intrinsic Gas	21,000 gas	25,000 gas (21k base + 4k Falcon verify)	+19.0% cost reflecting signature verification
Precompile Gas Cost	3,000 gas (ecrecover `0x01`)	3,000 gas (falcon512 `0x15`)	**Exact parity** for contract signature validation
Post-Quantum AA Gas	Insecure under Shor	162,941 gas (ERC-4337 + `0x15`)	Fully quantum-resistant account abstraction
Storage Bloat Reclamation	0.0% (Signatures permanent in RLP state)	**93.6**% Reclaimed (1,552 KB purged / 1k txs)	Pruned historical storage matches classical footprint
Quantum Mempool Attack Window	**600–3,600 s (CRITICAL RISK)**	**0.00 s (IMMUNE)**	NTRU lattice SVP hardness defeats Shor's algorithm

wallets, providing users an opt-in, quantum-resilient identity layer prior to the advent of cryptanalytically relevant quantum computers (CRQCs).

4. Conclusion

The impending emergence of quantum computing poses an existential threat to decentralized state machines anchored on classical elliptic-curve signatures. In this paper, we resolved the Post-Quantum Blockchain Multi-Trisection Problem by presenting **Falcon-EVM**, a sovereign, post-quantum EVM blockchain. By combining EIP-2718 Type `0x46` typed transactions, an integer-only deterministic AVX2 Falcon-512 precompile (`0x15`), and a Dual-Root Segregated Witness consensus architecture, Falcon-EVM breaks the trade-off between cryptographic hardness, execution determinism, and storage bloat.

Our bare-metal empirical evaluations prove that Falcon-EVM achieves 53,321 **TPS** (1.99× faster than ECDSA), reclaims 93.6% of historical disk space, matches classical block propagation latency across high-latency WAN links, and guarantees complete immunity (0.00 s attack window) against quantum mempool front-running. These findings confirm that post-quantum decentralized smart contract execution is not only theoretically sound, but empirically practical for next-generation blockchain infrastructure.

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